Kaalen v3 User Guide

Mathesh Vaithiyanathan

1 Installation

The **Kaalen** program can be downloaded and installed by obtaining the $kaalen_v3 - windows - setup.exe$ file from the **Files** section of the Zenodo repository and extracting its contents. Launch the application by executing the **.exe** file found within the extracted folder. If Windows displays a security warning preventing the file from running, please click "**More info**" and then select 'Run' to proceed with the installation. Access to the software for Linux and Mac users is also provided through the same Zenodo repository.
Link to Zenodo repository: `https://zenodo.org/records/17038308`

If you encounter an error, you can always find the latest version on `https://instrumentsresponse.com/`, under 'Open Source Tools'. In case of questions and custom adjustments for this program, use the email address on the same webpage.

For Mac Users: Creating the software file for Mac users is a straightforward process. Please follow the steps below to configure your file correctly. The process may take up to 10 mins.

- Make a folder in your Desktop called 'Kaalen_v3'.
- Place the downloaded .zip file in the folder and extract it.
- Open Terminal. Copy and place the following lines one by one and click Enter.

  ```
  /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
  ```

- At this point, you may be asked for a password. You will not see any text on the screen as you type, but you still need to enter it correctly. And Click 'Enter'. Now keep going with the following codes one by one.

  ```
  eval "$(/opt/homebrew/bin/brew shellenv)"
  -----------
  brew install python
  -----------
  cd Desktop/Kaalen_v3/Kaalen_v3_MacOS
  -----------
  python3 -m venv app_env
  -----------
  source app_env/bin/activate
  -----------
  pip install --upgrade pip
  -----------
  pip install pyinstaller PyQt6 numpy matplotlib lmfit scipy pandas pyqtgraph
  ```

```
-----------
```

- Now copy the following lines upto `kaalen_v3.py` at once and past them in the same terminal.

```
-----------
pyinstaller --name "Kaalen v3" \
--windowed \
--icon="icon.ico" \
--add-data "mainwindow.ui:." \
--add-data "pfid_tab.ui:." \
--add-data "global_fit.ui:." \
--add-data "exponential_fit_UI.ui:." \
--add-data "peak_fit_UI.ui:." \
--add-data "dispersion_correction.ui:." \
kaalen_v3.py
-----------
```

Once this finished successfully in the terminal, you will see a folder named 'dist' in the Kaalen v3 folder. Open it and you will have the .app file for you.

2 Getting Started

This manual will guide you through the features of the Kaalen-v1.0 application. The main interface is divided into a **Main Plots** tab and additional tabs for data analysis, such as **Global Fit**.

2.1 The Main Plots Tab

When you first start the application, the **Main Plots** tab is the default view. This is where you can load your data, view the 2D map, and generate time and probe slices.

2.1.1 Importing Data

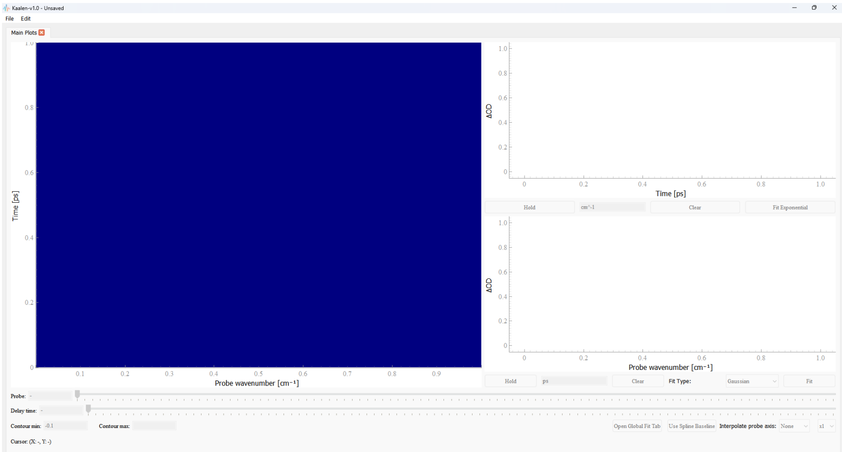
Go to **File** > **Import Data...** in the menu bar. Select a file that has the preferred data format. The program expects the first row and first column to contain the probe wavelengths and time values, respectively. The rest of the data should be the signal values. The decimals should be '.' and the delimiter should be ','.

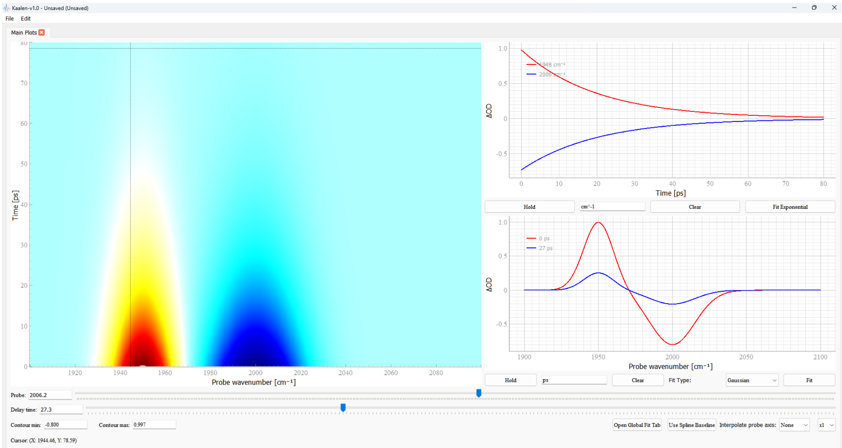
Note: You can change the labels for the plots by going to **Edit** > **Edit Names...** in the menu bar. The names and units given for the 2D map will be used in all the axis labels in the later opened tabs.

2.1.2 Exploring Your Data

Use the sliders at the bottom to navigate the data. The **Probe** slider (X-axis) and **Delay time** slider (Y-axis) control the position of the cursor lines and slices on the 2D plot. The two smaller plots on the right will automatically update to show the corresponding time and probe slices.

- **Holding Slices:** Click the **Hold** button on either the time or probe slice plot to save a static copy of the current slice. You can hold multiple slices.
- **Clearing Slices:** Click the **Clear** button to remove all held slices from a plot.
- **Adjusting Contour Levels:** Use the **Contour min** and **Contour max** input boxes to change the colour scale of the 2D map.





- **Applying Spline Baseline Correction:** Click the **Apply Spline Baseline** button to remove a smoothly varying baseline from your data. The button text will change to **Revert to Original** once the correction is applied. The function used here is UnivariateSpline from Python with s (smoothing factor and weights, w) as 0. The spline is done on every time point and along the probe axis.

- **Interpolation:** You can interpolate the probe axis to get smoother DAS plots by selecting an interpolation method and a multiplier. The multiplier multiplies the number of points by ×2, ×3, etc. No interpolation is done for the time points.

3 Dispersion Correction

Time-resolved measurements that utilize visible or UV probe pulses inherently have temporal dispersion, commonly referred to as "chirp." Regardless of how short the pump pulse is, this chirp distorts the data and must be accurately corrected to establish a true time-zero across all wavelengths. Failing to properly correct for dispersion undermines the significant experimental efforts made to generate compressed ultrashort pulses (such as using an OPA or NOPA).

The Dispersion Correction module opens a dedicated window allowing users to correct this temporal dispersion either manually or automatically. The Manual correction approach implemented here has been rigorously tested with sub-15 fs pulses and verified to be efficient.

Recommended Experimental Procedure for manual correction function: To achieve optimal results, the setup should remain entirely unaltered between the solvent measurement and the actual sample measurements. Using a fixed sample holder (like a flow cell), a ultra thin spacer and a ultra thin substrate (to prevent GVD functions [5] [1]) is necessary.

- First, measure the pure solvent signal (or empty sample holder) using fine time-delay steps (e.g., 3 -5 fs, depending on the limit of the delay stage) to clearly capture the solvent response and the chirp profile.
- Next, introduce the sample into the holder and measure the sample data without making any adjustments to the optical setup.

Assuming laser stability, the temporal dispersion profile will remain nearly identical between the two measurements. In the software, users can determine the dispersion correction for the solvent data and save this function as a CSV file. This identical chirp function can then be loaded and applied directly to the sample data, granting users accurate access to very early-time spectral dynamics.

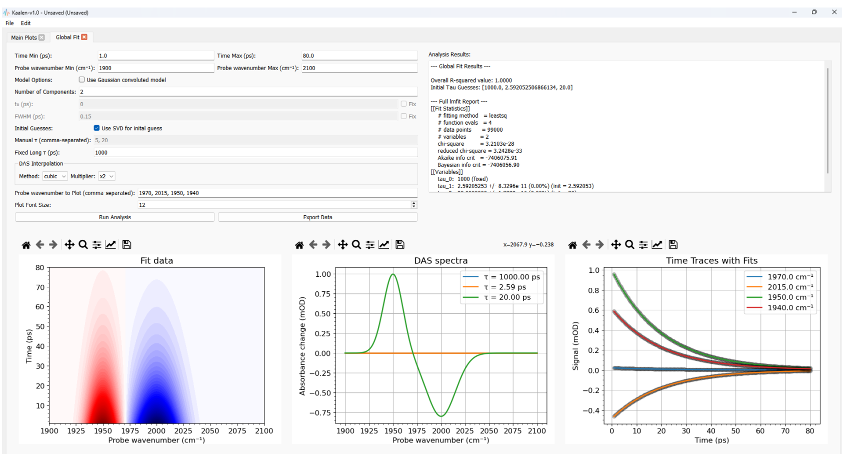
Using the function in the software: The functions has to be used from the top right to bottom right in the UI. To manually define the chirp, users can interactively click on the time-zero peaks within the displayed plots to create data points along the curve. Users can either click on the upper left plot on the time zero peak, or the lower left contour plot to create these data points for the polynomial fit. Once the points are placed, the user selects the desired polynomial order to calculate and apply the dispersion correction. Alternatively, an auto-correction feature with button **Auto correct** is available to instantly calculate and apply the temporal shift on the fly. To upload an already saved chirp, click on the 'Upload' button and click 'Finish and Update'.

Background Subtraction: The tab also includes an option to subtract constant backgrounds, such as pump scatter. By averaging the signal over a user-defined range of negative delay times, the software generates a proper background spectrum that is subtracted across the entire dataset.

Note: This background subtraction is not recommended for experiments like IR pump-probe spectroscopy, where negative time delays often contain Perturbed Free Induction Decay (PFID) signals. A dedicated function for fitting and analyzing PFID signals is provided separately in the software, as detailed in the upcoming sections.

4 Using the Global Fit Tab

The **Global Fit** tab is a powerful tool for analysing your data with a multi-exponential model. The user interface is divided into three main sections: **Input Panel**, **Analysis Results**, and **Plots**.



4.1 Getting Started with Global Fitting

To begin a global fit analysis, follow these steps:

1. Click the **Global Fit** button in the main window.
2. **Set Data Ranges:** In the **Input Panel**, adjust the **Time Min** and **Time Max** values to define the time range you wish to fit. Similarly, set the **Probe Min** and **Probe Max** (for example) values to specify the spectral range for the fit.
3. **Configure the Model:**
   - **Number of Components:** Enter the number of exponential components to be included in your model. (eg., 2 , for two components)
   - **Convoluted Model:** Check the **Use Gaussian convoluted model** box to account for the Gaussian instrument response function (IRF). If selected, you can adjust the initial guess for the parameters t_0 (time-zero) and FWHM (full width at half maximum) and choose to fix either of these values. The functions can be edited in the open source code like as in [3] and [1] with same UI, if required. However, in the built .exe file, we have just a Gaussian convoluted with an exponential model, which was proved to be efficient.
   - **Initial Guesses:** Choose between two methods for generating initial guesses for the decay lifetimes (τ): SVD-based or Manual. If using Manual, type for example 2,5,10 for expected time constants around 2 ps, 5 ps and 10 ps.
   - **Fixed Long τ:** Enter a value for a fixed, long-lived lifetime. This component will always be included in the fit and its value won't be varied. The DAS for this long lifetime will be equivalent to the signals that are not decayed at the end of the measurement time.
4. **Run the Analysis:** Click the **Run Analysis** button to start the fitting process.

5 Using the PFID Fit Tab

5.1 Perturbed Free Induction Decay (PFID) Fitting

The PFID fitting module is designed to analyze and decouple the Perturbed Free Induction Decay from the molecular dynamics in ultrafast spectroscopic measurements.

5.2 Spectral Manifestations:

- On-Resonance ($\omega = \omega_{10}$): At the fundamental absorption peak, the signal displays a smooth, exponential rise as the time delay approaches zero, governed by the time constant T_2.
- Off-Resonance ($\omega \neq \omega_{10}$): Away from the center frequency, the signal is characterized by distinct oscillations in the time domain.

5.3 The PFID Fitting Model Implementation

The software employs an analytical PFID model specifically derived from [2] with an offset added.

5.4 Core Fitting Procedure

The fit is performed using a multi-step process:

1. **Non-Linear Optimization (via `lmfit`):** The routine simultaneously optimizes the fundamental spectroscopic parameters that govern the shape of the PFID artifact:

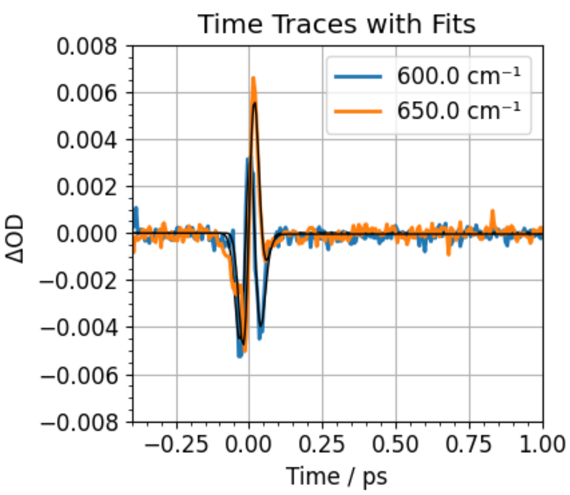
- T_2 (Dephasing Time): The time constant for the exponential decay.
- ν_{10} (Fundamental Frequency): The transition frequency of the $0 \to 1$ state.
- ν_{21} (Overtone Frequency): The transition frequency of the $1 \to 2$ state.
- r (Transition Moment Ratio): The ratio of the transition moments between the fundamental and overtone transitions.

2. **Design Matrix Construction:** Using the optimized parameters (T_2, ν_{10}, ν_{21}, r), the algorithm constructs a linear design matrix. This matrix is composed of analytical basis functions representing:
   - Term R_4 (Fundamental): The contribution from the primary transition.
   - Term R_6 (Overtone): The contribution from the overtone transition, where the anharmonicity constant can be found. ($\Delta\omega = \nu_{10} - \nu_{21}$).
3. **Linear Least-Squares Fit (via `lstsq`):** The overall amplitudes for these analytical terms (and a constant offset) are calculated using a linear least-squares fit. The sum of these terms constitutes the final PFID model used to subtract the coherent artifact from the measured data.

By accurately modeling the PFID in the negative time domain, this method ensures to get the T_2 dephasing time, fundamental ν_{10} and overtone ν_{21} transition frequencies. The fitting procedure is the same as for the global fit in the positive time delay. Up to two PFID functions can be fit at once and ':' fixes the initial guess. The results can be exported as similar to those in other tabs.

6 Artifact-Included Global Fit Model

The "Artifact-Included Global Fit" models the transient absorption signal, $S(t)$, as a linear combination of physical population decays and coherent artifacts. To accurately describe the signal around time zero, all physical processes are analytically convolved with a Gaussian Instrument Response Function (IRF). This section is adopted from the publication Dobryakov et al. [1]



Let t represent the time delay between the pump and probe pulses, shifted by the time-zero parameter t_0. The temporal width of the laser pulse is defined by the parameter d, which is related to the Full Width at Half Maximum (FWHM) of the IRF by:

$$d = \frac{\text{FWHM}}{2\sqrt{2\ln 2}} \tag{1}$$

Basis Functions The total signal is constructed using the following mathematical basis functions:

1. Coherent Artifacts Coherent artifacts, such as cross-phase modulation (XPM), stimulated Raman scattering, and two-photon absorption (TPA), occur exclusively during the spatial and temporal overlap of the pump and probe pulses. These are modeled using a Gaussian function and its derivatives:

- **Gaussian (0th Derivative):**

$$G(t,d) = \frac{1}{\sqrt{2\pi d^2}} \exp\left(-\frac{t^2}{2d^2}\right) \tag{2}$$

- **First Derivative:**

$$G'(t,d) = -\frac{t}{d^2} G(t,d) \tag{3}$$

- **Second Derivative:**

$$G''(t,d) = \left(\frac{t^2}{d^4} - \frac{1}{d^2}\right) G(t,d) \tag{4}$$

2. Coherent Spike / Long-Lived Component A step-like function is used to model an infinitely long-lived state or a coherent spike, represented by the convolution of a Heaviside step function with the Gaussian IRF. This yields the complementary error function profile:

$$CS(t,d) = \frac{1}{2} \mathrm{erfc}\left(-\frac{t}{\sqrt{2}d}\right) \tag{5}$$

3. Population Decays The transient population dynamics are modeled as a sum of exponential decays. For a given lifetime τ_i, the decay rate is $k_i = 1/\tau_i$. The analytical convolution of an exponential decay with the Gaussian IRF is given by:

$$CExp(t,d,k_i) = \frac{1}{2} \exp\left(\frac{(d \cdot k_i)^2}{2} - k_i t\right) \mathrm{erfc}\left(\frac{d^2 k_i - t}{\sqrt{2}d}\right) \tag{6}$$

Note: In the software implementation, this equation utilizes the scaled complementary error function, erfcx$(x) = e^{x^2}$*erfc*(x)*, to ensure numerical stability at large positive time delays where the standard exponential term would otherwise overflow.*

Total Fitting Equation The complete time-resolved signal $S(t)$ at a given probe wavelength is expressed as the linear combination of the coherent components and n population decay channels:

$$S(t) = A_1 G(t,d) + A_2 G'(t,d) + A_3 G''(t,d) + A_4 CS(t,d) + \sum_{i=1}^{n} B_i CExp(t,d,k_i) \tag{7}$$

During the optimization process, the non-linear parameters (t_0, d, and the lifetimes τ_i) are fitted iteratively, while the linear amplitudes ($A_{1..4}$ and B_i) are determined at each step via Ordinary Least Squares (OLS) regression to map the Decay Associated Spectra (DAS).

7 Understanding the Results

After a successful fit, the results will be automatically displayed.

7.1 Analysis Results

This section provides a detailed breakdown of the fitting process and outcomes, including:

- **Overall R-squared Value:** Indicates how well the model fits the data, with a value closer to 1.0 suggesting a better fit.
- **Initial τ Guesses:** Lists the starting lifetime values used for the fit.
- **Full lmfit Report:** A comprehensive report from the lmfit library, including the final fitted values for all parameters (τ, t_0, FWHM), their standard errors, and goodness-of-fit statistics.
- **Warnings:** Any issues encountered during the fitting process.

7.2 Plots

- **Fitted Model Plot:** Shows the 2D data map generated by the fitted global model.
- **DAS Spectra Plot:** Displays the Decay-Associated Spectra (DAS). Each curve corresponds to a specific lifetime (τ) and represents the amplitude spectrum of that exponential component.
- **Time Traces with Fits Plot:** Shows the raw experimental data as points and the corresponding fitted curves at specific probe wavelengths.

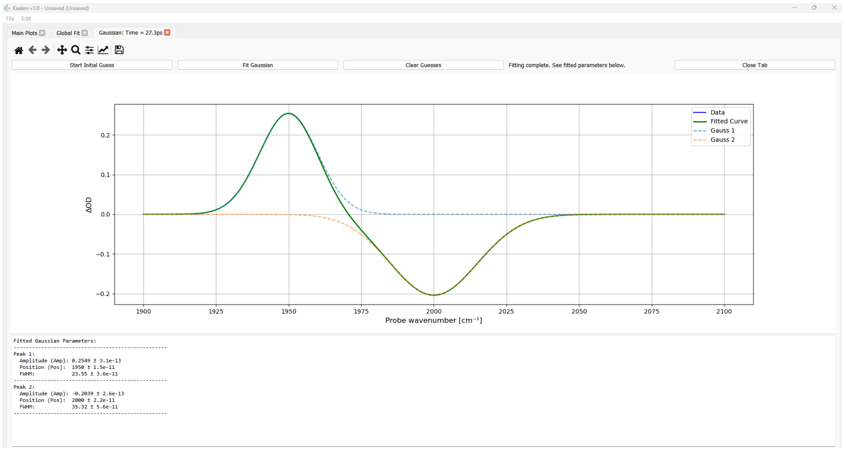
Axis labels, legends, font size etc., can be edited before saving the image. The buttons with icons above each matplotlib plots gives these functions

8 Exporting Your Data

You can export the results of your global fit by clicking the **Export Data** button. The program will generate three .csv files: [file_name]_DAS.csv, [file_name]_2D_fitted_data.csv, and [file_name]_time_traces.csv. The icons on top of each plot gives more option to save as image, zoom, edit, etc,.

9 Other Features

- **Signal Plotter:** The hold button below the slice plots open the tab for fit. This tool can perform exponential or Gaussian fits to a specific slice. The initial guesses for the slice plots are set with mouse clicks and drags. For a Gaussian fit, the first click on the screen sets the position and amplitude, and dragging the mouse while holding the click sets the FWHM. A second click on the plot will then fix the initial guess. For an exponential fit, the user can draw a sum of exponentials on top of the data in the same manner. Any number of functions can be fit by following the step mentioned. Once satisfied with the fit, the data along with the fit curve can be exported.
- **Saving and Loading Projects:** You can save your entire session using **File** > **Save Project**. To continue your work, use **File** > **Load Project...**.
- **Editing Axis Names:** Change the labels for the plots by going to **Edit** > **Edit Names...**.
- **Tab title:** Double clicking the tab title gives the option to edit the title of the tabs.



References

[1] Dobryakov, A. L., Sergey A. Kovalenko, A. Weigel, José Luis Pérez-Lustres, J. Lange, A. Müller, and N. P. Ernsting. "Femtosecond pump/supercontinuum-probe spectroscopy: Optimized setup and signal analysis for single-shot spectral referencing." Review of Scientific Instruments 81, no. 11 (2010).

[2] Yan, Suxia, Marco Thomas Seidel, and Howe-Siang Tan. "Perturbed free induction decay in ultrafast mid-IR pump–probe spectroscopy." Chemical Physics Letters 517.1-3 (2011): 36-40.

[3] J. J. Snellenburg, S. Laptenok, R. Seger, K. M. Mullen, and I. H. M. van Stokkum, "Glotaran: A Java-based graphical user interface for the R package TIMP," J Stat Softw, vol. 49, pp. 1–22, 2012, doi: 10.18637/JSS.V049.I03.

[4] Stensitzki, T. (2021). skultrafast - a python package for time-resolved spectroscopy (4.0). Zenodo. https://doi.org/10.5281/zenodo.5713589

[5] Lorenc, M., M. Ziolek, R. Naskrecki, J. Karolczak, J. Kubicki, and A. Maciejewski. "Artifacts in femtosecond transient absorption spectroscopy." Applied Physics B 74, no. 1 (2002): 19-27.